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Role: Backend (Register and Login) Creator and Documentation Lead

My main task is on the server-side where I developed the backend API of our system's user authentication module. I am responsible for the logic user registration and login validation. I also helped in the documentation and divided the tasks and assigned roles to the group members.

First, I developed the authentication system using PHP and MySQL. I created a single endpoint for both user registration and login within a single file. I handled incoming requests using php://input and $POST. I used an action parameter to decide if the request id for registration or the login. The two logics now exist in one centralized endpoint.

For registration, I made the input validation that ensures that all the required fields are filled. I also added checks to prevent duplicate usernames and email addresses by checking the database first. Passwords are securely stored using hashing instead of plain-text. If everything is valid, the user data is saved to the database and a JSON response is returned with the correct HTTP status code. For the login, I checked if the email exists in the database and verified the password using password_verify(). If the login is successful, user detail is returned in a structured JSON response. I ensured that all responses follow a structured JSON format with status, code, message. I also made sure to properly show an error in JSON format when the database connection fails, so it is clear when the system cannot connect to the database.

However, I encountered several challenges during development. One of the main issues was combining the separate PHP files for authentication and seller functions into one file where there was an issue in variables and not being able to connect the two files at first. I also experienced common mistakes like missing commas, incorrect operators, typos, and mismatch variables. I also often forget to set HTTP response codes in some parts of the code. Since the API is centralized, a single and small error in one section often affected the other parts.

To solve these issues, I improved the routing by properly organizing the action parameter so that authentication runs first, while other functions are handled separately using switch structure. This allowed both authentication and seller functionalities to work together without conflict. I also carefully checked variable names throughout the code to make them consistent. Lastly, I always make sure to validate HTTP response codes for every response to that debugging and the responses are clear.